Submission Instructions for Assignment 2

ACL-style LaTeX Report:

You can compile the ACL LaTeX template using Overleaf.

https://www.overleaf.com/latex/templates/association-for-computational-linguistics-acl-conference/jvxskxpnznfj

Jupyter Notebook:

Please submit a notebook containing the experimental details in the Playground section.

Result Files:

We have prepared functions in the notebook for saving your results to local files. These files will be used to assess your experimental results. Please follow these steps to prepare them:

1. Task 1: Once the prompting results are successfully generated, save the zero-shot prompting results by running the following code block provided in the notebook.

   Export the results for uploading. Please do not modifiy the output format!

   ```
   [ ] def save_results(dataset, pred, output_file):
           predicted_texts = zip(dataset, pred)
           # Write the paired results to a text file
           with open(output_file, "w") as f:
               for text, prediction in predicted_texts:
                   f.write(f"{prediction} ||| {text}\n")

       save_results(sst5_data["validation"]["text"], sst5_val_pred, "sst5-val-prompting.txt")
       save_results(sst5_data["test"]["text"], sst5_test_pred, "sst5-test-prompting.txt")
       save_results(rto_data["validation"]["text"], rto_val_pred, "rto-val-prompting.txt")
       save_results(rto_data["test"]["text"], rto_test_pred, "rto-test-prompting.txt")
   ```

   You may get the following files:

   - `rto-test-prompting.txt`: Classification results of zero-shot prompting for "Rotten Tomatoes" test data.
   - `rto-val-prompting.txt`: Classification results of zero-shot prompting for "Rotten Tomatoes" validation data.
   - `sst5-test-prompting.txt`: Classification results of zero-shot prompting for "SST5" test data.
   - `sst5-val-prompting.txt`: Classification results of zero-shot prompting for "SST5" validation data.

2. Task 2: Complete the evaluation code and run the following code block provided in the notebook:

Export the results for uploading. Please do not modifiy the output format!

```
results = []

# RTO Data Validation
rto_val_accuracy = calculate_accuracy(
    y_true=rto_data["validation"]["label"], y_pred=rto_val_pred
)
rto_val_f1 = calculate_f1_score(
    y_true=rto_data["validation"]["label"], y_pred=rto_val_pred
)
results.append(f"RTO Validation - Accuracy: {rto_val_accuracy}, F1 Score: {rto_val_f1}")

# RTO Data Test
rto_test_accuracy = calculate_accuracy(
    y_true=rto_data["test"]["label"], y_pred=rto_test_pred
)
rto_test_f1 = calculate_f1_score(y_true=rto_data["test"]["label"], y_pred=rto_test_pred)
results.append(f"RTO Test - Accuracy: {rto_test_accuracy}, F1 Score: {rto_test_f1}")

# SST5 Data Validation
sst5_val_accuracy = calculate_accuracy(
    y_true=sst5_data["validation"]["label"], y_pred=sst5_val_pred
)
sst5_val_f1 = calculate_f1_score(
    y_true=sst5_data["validation"]["label"], y_pred=sst5_val_pred, average="macro"
)
results.append(
    f"SST5 Validation - Accuracy: {sst5_val_accuracy}, F1 Score: {sst5_val_f1}"
)

# SST5 Data Test
sst5_test_accuracy = calculate_accuracy(
    y_true=sst5_data["test"]["label"], y_pred=sst5_test_pred
)
sst5_test_f1 = calculate_f1_score(
    y_true=sst5_data["test"]["label"], y_pred=sst5_test_pred, average="macro"
)
results.append(f"SST5 Test - Accuracy: {sst5_test_accuracy}, F1 Score: {sst5_test_f1}")

# Print results
for result in results:
    print(result)

# Write results to a text file
with open("task1-scores.txt", "w") as f:
    for result in results:
        f.write(result + "\n")
```

You may get the following file:

- `task1-scores.txt`

3. Task 3: After fine-tuning the model, evaluate your tuned model on the previous datasets and save the results by running the following code block provided in the notebook:

Export the results for uploading. Please do not modifiy the output format!

```python
results = []

# RTO Data Validation
rto_val_accuracy = calculate_accuracy(y_true=rto_data["validation"]["label"], y_pred=tuned_rto_val_pred)
rto_val_f1 = calculate_f1_score(y_true=rto_data["validation"]["label"], y_pred=tuned_rto_val_pred)
results.append(f"RTO Validation - Accuracy: {rto_val_accuracy}, F1 Score: {rto_val_f1}")

# RTO Data Test
rto_test_accuracy = calculate_accuracy(y_true=rto_data["test"]["label"], y_pred=tuned_rto_test_pred)
rto_test_f1 = calculate_f1_score(y_true=rto_data["test"]["label"], y_pred=tuned_rto_test_pred)
results.append(f"RTO Test - Accuracy: {rto_test_accuracy}, F1 Score: {rto_test_f1}")

# SST5 Data Validation
sst5_val_accuracy = calculate_accuracy(y_true=sst5_data["validation"]["label"], y_pred=tuned_sst5_val_pred)
sst5_val_f1 = calculate_f1_score(y_true=sst5_data["validation"]["label"], y_pred=tuned_sst5_val_pred, average="macro")
results.append(f"SST5 Validation - Accuracy: {sst5_val_accuracy}, F1 Score: {sst5_val_f1}")

# SST5 Data Test
sst5_test_accuracy = calculate_accuracy(y_true=sst5_data["test"]["label"], y_pred=tuned_sst5_test_pred)
sst5_test_f1 = calculate_f1_score(y_true=sst5_data["test"]["label"], y_pred=tuned_sst5_test_pred, average="macro")
results.append(f"SST5 Test - Accuracy: {sst5_test_accuracy}, F1 Score: {sst5_test_f1}")

# Print results
for result in results:
    print(result)

# Write results to a text file
with open('task3-scores.txt', 'w') as f:
    for result in results:
        f.write(result + '\n')
```

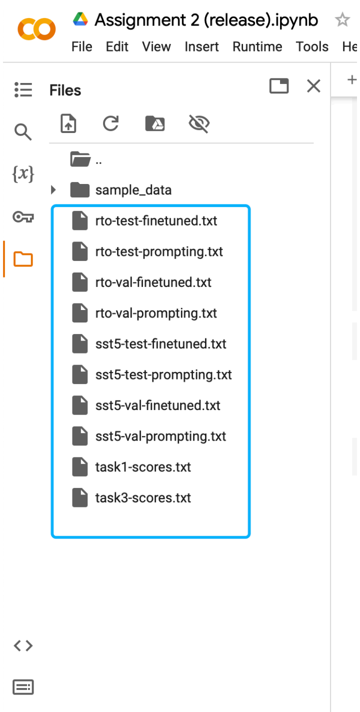
```python
def save_results(dataset, pred, output_file):
    predicted_texts = zip(dataset, pred)
    # Write the paired results to a text file
    with open(output_file, "w") as f:
        for text, prediction in predicted_texts:
            f.write(f"{prediction} ||| {text}\n")

save_results(sst5_data["validation"]["text"], tuned_sst5_val_pred, "sst5-val-finetuned.txt")
save_results(sst5_data["test"]["text"], tuned_sst5_test_pred, "sst5-test-finetuned.txt")
save_results(rto_data["validation"]["text"], tuned_rto_val_pred, "rto-val-finetuned.txt")
save_results(rto_data["test"]["text"], tuned_rto_test_pred, "rto-test-finetuned.txt")
```

You may get the following files:

- `rto-test-finetuned.txt`: Classification results of the fine-tuned model for "Rotten Tomatoes" test data.
- `rto-val-finetuned.txt`: Classification results of the fine-tuned model for "Rotten Tomatoes" validation data.
- `sst5-test-finetuned.txt`: Classification results of the fine-tuned model for "SST5" test data.
- `sst5-val-finetuned.txt`: Classification results of the fine-tuned model for "SST5" validation data.
- `task3-scores.txt`: Evaluation results from Task 3.

4. File Submission: Do not rename the files. Upload these files along with your report and notebook to UMMoodle. If you are using an online GPU environment, be sure to download them from the temporary workspace (see screenshot below).


Assignment 2 (release).ipynb
File
Edit
View
Insert
Runtime
Tools
He
Files
..
sample_data
rto-test-finetuned.txt
rto-test-prompting.txt
rto-val-finetuned.txt
rto-val-prompting.txt
sst5-test-finetuned.txt
sst5-test-prompting.txt
sst5-val-finetuned.txt
sst5-val-prompting.txt
task1-scores.txt
task3-scores.txt